

ELITE MATHEMATICS

Pearson Edexcel International GCSE

Linear 4MA1

Adaptive Classified

OL-T38 · Differentiation

With Answers

Questions with full worked solutions

■ 36 pages

FAMILY MAP

Differentiation

F01	Find and classify stationary points	3 Q 17 marks
F02	Use the derivative as gradient or rate	6 Q 29 marks
F03	Apply differentiation to linear kinematics	4 Q 19 marks
F04	Optimise a practical model	3 Q 15 marks

17 $y = 4x^3 + 5x^2 + 2x$

(a) Find $\frac{dy}{dx}$

$$\frac{dy}{dx} = \dots\dots\dots (2)$$

(b) Find the coordinates of the turning points on the graph with equation $y = 4x^3 + 5x^2 + 2x$
Show clear algebraic working.

$$\dots\dots\dots (4)$$

(Total for Question 17 is 6 marks)

Worked Solution 28

Derivative and turning points

UNIT 2
6 marks**Differentiate**

$$\frac{dy}{dx} = 12x^2 + 10x + 2.$$

Solve derivative equal to zero

$$12x^2 + 10x + 2 = 2(3x + 1)(2x + 1) = 0 \Rightarrow x = -\frac{1}{3}, -\frac{1}{2}.$$

Substitute

$$x = -\frac{1}{2} \Rightarrow y = -\frac{1}{4}; \quad x = -\frac{1}{3} \Rightarrow y = -\frac{7}{27}.$$

Final answer

$$\frac{dy}{dx} = 12x^2 + 10x + 2; \quad \left(-\frac{1}{2}, -\frac{1}{4}\right), \left(-\frac{1}{3}, -\frac{7}{27}\right)$$

Question 08

4MA1-1H Q18

ORIGINAL QUESTION

5 marks

18 A curve C has equation $y = x^3 - 40x + 1$

Find the coordinates of both the points on C at which the gradient is 8

(..... ,)

(..... ,)

Worked Solution 08

Gradient on a curve

MODULAR UNIT 2

5 marks

Differentiate

$$dy/dx = 3x^2 - 40.$$

Set the gradient

$$3x^2 - 40 = 8 \Rightarrow x = \pm 4.$$

Coordinates

$$y(4) = -95, \quad y(-4) = 97.$$

Final answer

$$(4, -95), (-4, 97)$$

Question 13

4MA1-1H Q16 demand

ORIGINAL QUESTION UNIT 2

4 marks

16 A curve has equation $y = 4x^3 - 8x + 5$

Find the x coordinates of the two points on the curve where the gradient is $\frac{1}{3}$

.....

Worked solution 13

Coordinates with a given gradient

MODULAR UNIT 2

4 marks

Demand – Coordinates with a given gradient

Differentiate

$$\frac{dy}{dx} = 12x^2 - 8.$$

Set the gradient

$$12x^2 - 8 = \frac{1}{3} \Rightarrow 12x^2 = \frac{25}{3} \Rightarrow x^2 = \frac{25}{36}.$$

Take both roots

$$x = \pm \frac{5}{6}.$$

Final answer

$$x = \pm 5/6$$

Question 30

4MA1-2H Q18 whole

ORIGINAL QUESTION UNIT 2

4 marks

- 18 A particle is moving along a straight line that passes through the fixed point O
The displacement, s metres, of the particle from O at time t seconds is given by

$$s = 2t^3 - 5t^2 + 6t - 5$$

Find the value of t when the acceleration of the particle is 5 m/s^2

$t = \dots\dots\dots$

Worked solution 30

Acceleration from a displacement function

MODULAR UNIT 2

4 marks

whole – Acceleration from a displacement function

Differentiate once

$$v = \frac{ds}{dt} = 6t^2 - 10t + 6.$$

Differentiate again

$$a = \frac{dv}{dt} = 12t - 10.$$

Set the required acceleration

$$12t - 10 = 5 \Rightarrow 12t = 15.$$

Solve

$$t = \frac{15}{12} = 1.25 \text{ s.}$$

Final answer

t = 1.25 s

Question 11

4MA1-1H Q11

ORIGINAL QUESTION

6 marks

11 The curve **C** has equation $y = 4x^3 + x^2 - 20x$

(a) Find $\frac{dy}{dx}$

$$\frac{dy}{dx} = \dots\dots\dots$$

(2)

(b) Find the x coordinates of the points on **C** where the gradient is 4
Show clear algebraic working.

$$\dots\dots\dots$$

(4)

Worked Solution 11

Differentiate and solve a gradient condition

MODULAR UNIT 2

6 marks

Differentiate

$$\frac{dy}{dx} = 12x^2 + 2x - 20.$$

Set the gradient to 4

$$12x^2 + 2x - 20 = 4 \Rightarrow 6x^2 + x - 12 = 0.$$

Factorise

$$6x^2 + x - 12 = (3x - 4)(2x + 3).$$

Final answer

$$x = \frac{4}{3} \text{ or } x = -\frac{3}{2}$$

Original question 13

4MA1-1H Q20 M01

MODULAR UNIT 2

4 marks

20 The curve with equation $y = 2x^4 - 64x$ has a minimum point.

Find an equation of the tangent to the curve at the minimum point.

Show clear algebraic working.

.....

Worked solution 13

Find the tangent at a minimum

MODULAR UNIT 2

4 marks

M01 – Find the tangent at a minimum**Differentiate**

$$\frac{dy}{dx} = 8x^3 - 64.$$

Set the gradient to zero

$$8x^3 - 64 = 0 \Rightarrow x = 2.$$

Find the y-coordinate

$$y = 2(2^4) - 64(2) = -96.$$

State the horizontal tangent

$$y = -96.$$

Final answer

$$y = -96$$

Original question 15

4MA1-1H Q17 Q17

ORIGINAL QUESTION**5 marks**

- 17 A particle P moves along a straight line.
The fixed point O lies on this line.

The displacement of P from O at time t seconds, $t \geq 1$, is s metres where

$$s = 4t^2 + \frac{125}{t}$$

The velocity of P at time t seconds, $t \geq 1$, is v m/s

Work out the distance of P from O at the instant when $v = 0$

..... m

Worked solution

Family: Apply differentiation to linear kinematics • Variant: Instantaneous rest or direction reversal

METHOD / WORKING

MODULAR UNIT 2 • JUNE 2022 DETAILED SOLUTIONS
Worked solution 15 MODULAR UNIT 2
Chapter: Differentiation 5 marks
Q17 – Chapter: Differentiation
Family: Apply differentiation to linear kinematics • Variant: Instantaneous rest or direction reversal
Method
$s = 4t^2 + 125t^{-1}$, so $v = ds/dt$.
Why: Velocity is the derivative of displacement with respect to time.
Method
$v = 8t - 125t^{-2} = 8t - 125/t^2$.
Why: Differentiate each power of t .
Method
When $v = 0$, $8t - 125/t^2 = 0$, so $8t^3 = 125$ and $t = 2.5$ seconds.
Why: Multiply by t^2 and solve the resulting cubic for $t > 0$.
Method
At this time, $s = 4(2.5)^2 + 125/2.5$.
Why: Substitute the stationary time into the original displacement expression.
Method
$s = 25 + 50 = 75$, so the distance from O is 75 m.
Why: The displacement is positive, so its magnitude is the distance.
Final answer
distance: 75 m

FINAL ANSWER

Original question 15

4MA1-1H Q24

ORIGINAL QUESTION**6 marks**

24 The curve **C** has equation $y = ax^3 + bx^2 - 12x + 6$ where a and b are constants.

The point A with coordinates $(2, -6)$ lies on **C**

The gradient of the curve at A is 16

Find the y coordinate of the point on the curve whose x coordinate is 3

Show clear algebraic working.

$y = \dots\dots\dots$

Worked solution 15

Chapter: Differentiation

MODULAR UNIT 2

6 marks

Q24 – Chapter: Differentiation

Family: Find and classify stationary points • Variant: Recover curve parameters from point and gradient data

Method

At $(2, -6)$, $-6 = 8a + 4b - 24 + 6$, so $8a + 4b = 12$.

Why: Substitute the point into the cubic equation.

Method

$\frac{dy}{dx} = 3ax^2 + 2bx - 12$.

Why: Differentiate the curve.

Method

At $x=2$, $16 = 12a + 4b - 12$, so $12a + 4b = 28$.

Why: Use the given gradient at A.

Method

Solving $8a + 4b = 12$ and $12a + 4b = 28$ gives $a = 4$ and $b = -5$.

Why: Solve the simultaneous linear equations.

Method

At $x=3$, $y = 4(3)^3 - 5(3)^2 - 12(3) + 6$.

Why: Substitute $x=3$ and the constants.

Method

$y = 33$.

Why: Evaluate the expression.

Final answer

y: 33

Original question 35

4MA1-2H Q24

ORIGINAL QUESTION**5 marks**

24 A particle P moves along a straight line that passes through the fixed point O

The displacement, x metres, of P from O at time t seconds, where $t \geq 0$, is given by

$$x = 4t^3 - 27t + 8$$

The direction of motion of P reverses when P is at the point A on the line.

The acceleration of P at the instant when P is at A is $a \text{ m/s}^2$

Find the value of a

$a = \dots\dots\dots$

Worked solution

Family: Apply differentiation to linear kinematics • Variant: Instantaneous rest or direction reversal

METHOD / WORKING

MODULAR UNIT 2 • JANUARY 2022 DETAILED SOLUTIONS
Worked solution 35 MODULAR UNIT 2
Chapter: Differentiation 5 marks
Q24 – Chapter: Differentiation
Family: Apply differentiation to linear kinematics • Variant: Instantaneous rest or direction reversal
Method
$v = dx/dt = 12t^2 - 27$.
Why: Differentiate $x = 4t^3 - 27t + 8$.
Method
At a reversal $v = 0$, so $12t^2 - 27 = 0$ and $t^2 = 9/4$.
Why: Set velocity equal to zero.
Method
Since $t \geq 0$, $t = 3/2$.
Why: Choose the physical time.
Method
$a = dv/dt = 24t$.
Why: Differentiate velocity.
Method
At $t = 3/2$, $a = 36 \text{ m/s}^2$.
Why: Evaluate the acceleration.
Final answer
<i>acceleration : 36 m/s^2</i>

FINAL ANSWER

15

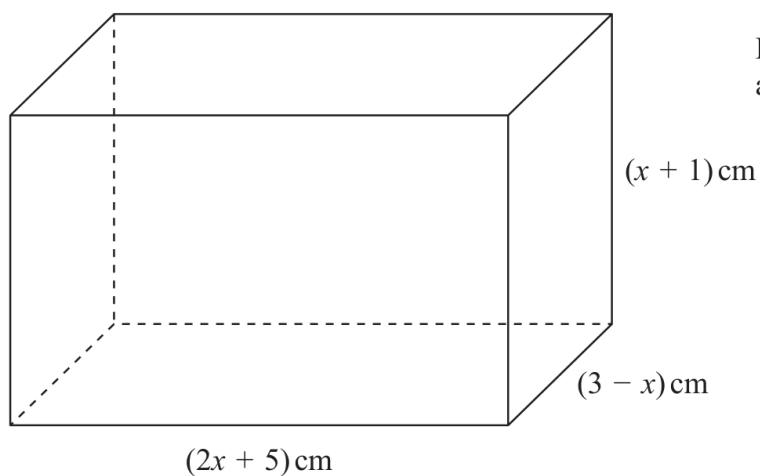


Diagram **NOT**
accurately drawn

The diagram shows a cuboid of volume $V \text{ cm}^3$

(a) Show that $V = 15 + 16x - x^2 - 2x^3$

There is a value of x for which the volume of the cuboid is a maximum.

(b) Find this value of x .
Show your working clearly.
Give your answer correct to 3 significant figures.

$x = \dots\dots\dots$
(5)

Worked solution

January 2020 | Question 15 | 5 marks

Family: Optimise a practical model | Variant: Maximum volume from geometric dimensions

METHOD / WORKING

1. $dV/dx = 16 - 2x - 6x^2$.
2. Set $dV/dx = 0 : 3x^2 + x - 8 = 0$.
3. Quadratic formula gives $x = (-1 \pm \sqrt{97})/6$.
4. The admissible positive root is 1.4748...
5. Hence $x = 1.47$ to 3 s.f.

FINAL ANSWER

 $x = 1.47$

17 A solid, **S**, is made from a hemisphere and a cylinder.

The centre of the circular face of the hemisphere and the centre of the top face of the cylinder are at the same point.

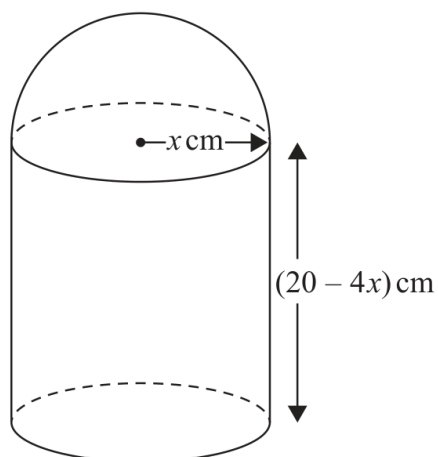


Diagram **NOT**
accurately drawn

The radius of the cylinder and the radius of the hemisphere are both $x \text{ cm}$.
The height of the cylinder is $(20 - 4x) \text{ cm}$.

The volume of **S** is $V \text{ cm}^3$ where $V = \frac{1}{3} \pi y$

Find the maximum value of y .
Show clear algebraic working.

Worked solution

January 2021 | Question 17 | 5 marks

Family: Optimise a practical model | Variant: Maximum volume from geometric dimensions

METHOD / WORKING

1. $\text{Volume } V = \frac{1}{3}\pi r^2 h = \frac{1}{3}\pi [(2/3)x^3 + x^2(20 - 4x)]$.
2. $\text{Since } V = \frac{1}{3}\pi y, y = 60x^2 - 10x^3$.
3. $\text{Differentiate : } dy/dx = 120x - 30x^2 \text{ and set equal to zero.}$
4. $x=4$ is the valid stationary value.
5. Substitution gives $y=320$, the maximum.

FINAL ANSWER

 $y=320$

- 19 A particle P is moving along a straight line.
The fixed point O lies on this line.

At time t seconds where $t \geq 0$, the displacement, s metres, of P from O is given by

$$s = t^3 + 5t^2 - 8t + 10$$

Find the displacement of P from O when P is instantaneously at rest.

Give your answer in the form $\frac{a}{b}$ where a and b are integers.

..... metres

Worked solution

January 2021 | Question 19 | 5 marks

Family: Apply differentiation to linear kinematics | Variant: Instantaneous rest or direction reversal

METHOD / WORKING

1. *Velocity* $v = ds/dt = 3t^2 + 10t - 8$.
2. Set $v=0$: $(3t-2)(t+4)=0$, so $t=2/3$ for $t \geq 0$.
3. *Substitute* $t = 2/3$ into $s = t^3 + 5t^2 - 8t + 10$.
4. Simplify the exact value.
5. Displacement = $194/27$ m.

FINAL ANSWER

194/27 m

- 21 The point A is the only stationary point on the curve with equation $y = kx^2 + \frac{16}{x}$ where k is a constant.

Given that the coordinates of A are $\left(\frac{2}{3}, a\right)$

find the value of a .

Show your working clearly.

$a = \dots\dots\dots$

Worked solution

June 2021 | Question 21 | 5 marks

Family: Find and classify stationary points | Variant: Recover curve parameters from point and gradient data

METHOD / WORKING

1. Differentiate $y = kx^2 + 16/x$: $dy/dx = 2kx - 16/x^2$.2. At the stationary point $x=2/3$, set $2k(2/3) - 16/(2/3)^2 = 0$, giving $k=27$.3. Substitute $x = 2/3$ into $y = 27x^2 + 16/x$.4. Evaluate $y = 27(4/9) + 16/(2/3) = 12 + 24 = 36$.5. Therefore $a=36$.

FINAL ANSWER

 $a=36$

23 Curve **C** has equation $y = px^3 - mx$ where p and m are positive integers.

Find the range of values of x , in terms of p and m , for which the gradient of **C** is negative.

(Total for Question 23 is 4 marks)

Answer reference | May-Nov 2020 | Question 23

Family: Use the derivative as gradient or rate • Variant: Points where gradient equals a value

23	$\left(\frac{dy}{dx} = 3px^2 - m\right)$		4	M1 for $3px^2$ or $-m$
	$3px^2 - m < 0$ oe			M1 ft dep on M1 for setting up an inequality with their ' $3px^2$ ' - ' m ' must be a two-term expression in the form $apx^2 \pm m$
	$\pm \sqrt{\frac{m}{3p}}$			B1 for both critical values
		$-\sqrt{\frac{m}{3p}} < x < \sqrt{\frac{m}{3p}}$		A1 may be seen as two separate inequalities
Total 4 marks				

Worked solution

May-Nov 2020 | Question 23 | 4 marks

Family: Use the derivative as gradient or rate | Variant: Points where gradient equals a value

METHOD / WORKING

Worked solution
May-Nov 2020 Question 23 4 marks
Family: Use the derivative as gradient or rate Variant: Points where gradient equals a value
Method
$y = px^3 - mx$
Differentiate:
$(dy)/(dx) = 3px^2 - m$
The gradient is negative when
$3px^2 - m < 0$
$3px^2 < m$
$x^2 < (m)/(3p)$
So
$-\sqrt{(m)/(3p)} < x < \sqrt{(m)/(3p)}$
Answer: $-\sqrt[3]{\frac{m}{3p}} < x < \sqrt[3]{\frac{m}{3p}}$

FINAL ANSWER

20 The radius of a right circular cylinder is x cm.

The height of the cylinder is $\left(\frac{800}{\pi x} - x\right)$ cm.

The volume of the cylinder is V cm³

Find the maximum value of V

Give your answer correct to the nearest whole number.



(Total for Question 20 is 5 marks)

Answer reference | January 2023 | Question 20

Family: Optimise a practical model • Variant: Maximum volume from geometric dimensions

20	$(V =) \pi x^2 \left(\frac{800}{\pi x} - x \right)$ or $800x - \pi x^3$ oe		5	M1 for volume of cylinder (in terms of one variable, e.g. x or r)
	$\left(\frac{dV}{dx} = \right) 800 - 3\pi x^2$			M1ft for differentiating an expression in one variable to find 800 or $\pm 3\pi x^2$ (must come from a cubic in the form $800x \pm \pi x^3$ or $800x \pm ax^3$ or $bx \pm \pi x^3$ where $a \neq 0$ and $b \neq 0$)
	" $800 - 3\pi x^2 = 0$ "			M1ft dep on previous M1 for equating their $\frac{dV}{dx}$ to zero (must be a quadratic in the form $800 \pm ax^2$ or $b \pm 3\pi x^2$ where $a \neq 0$ and $b \neq 0$)
	$(x =) \sqrt{\frac{800}{3\pi}} (= \sqrt{84.8(8263...)})$ or $\frac{\sqrt{9600\pi}}{6\pi}$ or $9.2(13177...)$			A1 for a correct value of x Allow use of quadratic formula
	Award marks within the range from correct working	4914		A1 accept 4910 – 4914
				Total 5 marks

Worked solution

January 2023 | Question 20 | 5 marks

Family: Optimise a practical model • Variant: Maximum volume from geometric dimensions

Official final mark-scheme pages are embedded immediately after this question.

22 Curve **C** has equation $y = x^3 - 16x + 7$

At two points on **C**, the gradient is 11

The tangents to **C** at these two points have equations of the form $y = ax + b$

Work out the two possible values of b

Show clear algebraic working.

(Total for Question 22 is 6 marks)

Worked solution

November 2025 | Question 22 | 6 marks

Family: Use the derivative as gradient or rate | Variant: Equation of a calculus tangent

METHOD / WORKING

Worked solution
November 2025 Question 22 6 marks
Family: Use the derivative as gradient or rate Variant: Equation of a calculus tangent
Method
$y = x^3 - 16x + 7$
Differentiate:
$(dy)/(dx) = 3x^2 - 16$
The gradient of the tangent is 11, so
$3x^2 - 16 = 11$
$3x^2 = 27$
$x^2 = 9$
$x=3$ quad or quad $x=-3$
When $x=3$:
$y=3^3-16(3)+7=27-48+7=-14$
The tangent has equation $y=11x+b$, so
$-14=11(3)+b$
$b=-47$
When $x=-3$:
$y=(-3)^3-16(-3)+7=-27+48+7=28$
$28=11(-3)+b$
$b=61$
Answer: boxed{b=-47 or b=61}
PROVISIONAL WORKED SOLUTION — NOVEMBER 2025 QP-ONLY EXCEPTION

FINAL ANSWER